

Lisha Maharjan

513-593-8706 | Oxford, Ohio | lishamaharjan358@gmail.com | [linkedin.com/in/lishamaharjan](https://www.linkedin.com/in/lishamaharjan)

SUMMARY

Detail-oriented Computer Science graduate with strong analytical and problem-solving skills. Proven track experience as a software engineer focused on data analytics and engineering building ETL pipelines, automating and optimizing data workflows, with a solid understanding of data engineering concepts. Passionate about leveraging technology to solve real-world challenges, continuously learning new tools, and driving innovation through data-driven insights.

EDUCATION

<i>Miami University, Ohio</i>	<i>Aug 2023 - Aug 2025</i>
Master of Science in Computer Science	GPA: 3.85/4
<i>Tribhuvan University, Nepal</i>	<i>Nov 2017 - Apr 2022</i>
Bachelor of Engineering in Computer Engineering	GPA: 3.73/4

RESEARCH EXPERIENCE

AI Prompt Engineering for Web Accessibility Issues , Miami University, Ohio	<i>Aug 2023 - Aug 2025</i>
- Defined research scope and objectives to investigate how OpenAI model can assist developers in addressing web accessibility issues. - Automated data extraction and preprocessing using Python to curate accessibility-related posts from StackOverflow database using keyword and tag-based search and pull requests with accessibility labels from GitHub repositories using Github API. - Constructed a filtered dataset of accessibility-related code changes by manually tagging the data to identify types of accessibility issues. - Designed a pipeline to query OpenAI model in Python, using prompt engineering to generate model responses based on real-world developer queries. - Applied prompt engineering to analyze raw code files and predict potential accessibility issues.	

WORK EXPERIENCE

Graduate Assistant , Miami University, Ohio	<i>Aug 2023 - Aug 2025</i>
- Assisted students in software architecture projects by implementing MVC design architecture in a monolithic game codebase, which improved modularity, modifiability, and maintainability. - Guided 40+ students in data structures, abstraction, and OOP (Object-oriented programming) concepts in Java, which enhanced their understanding of programming principles, test cases, and best practices on style standards. - Assessed UI/UX coursework projects focused on user-centered design processes, user requirements, and iterative prototyping, leading to improved usability and accessibility in student projects.	
Software Engineer , LIS, Nepal	<i>Apr 2022 - Jul 2023</i>
- Optimized SQL scripts for ETL pipelines for US retail giants, collaborating closely with business stakeholders to align data workflows with business requirements, enabling improved decision-making through timely reporting. - Streamlined data workflows to clean, transform, and aggregate large-scale retail datasets in Snowflake, enabling dynamic sales trend visualizations in Tableau with time-based filters (day/week/month/year) across multiple locations. - Developed a decision engine using FastAPI in Python and deployed using Docker to identify optimal store locations for order fulfillment by aggregating key operational dataset variables, improving order fulfillment process efficiency. - Built custom Azure Power App integrated with Power BI to automate healthcare inventory management by fetching scanned product details, eliminating manual Excel updates, and enabling real-time reporting for improved operational efficiency. - Enhanced system monitoring and error analysis by implementing ELK (Elastic, Logstash, and Kibana) architecture by parsing raw log files and visualizing log errors, increasing issue detection efficiency.	

- Developed machine learning anomaly detection model using unsupervised learning methods like **K-Means Clustering** and **Principal Component Analysis (PCA)** to group patterns into clusters and identify the anomalous error logs.

Software Engineer Intern, LIS, Nepal

Nov 2021 - Feb 2022

- Designed and developed retail **data warehouse** using **Snowflake** schema, creating **Python** scripts for **ETL** processes, streamlining data transfer from multiple source tables to Snowflake's target tables, and ensuring high data integrity.
- Built **Power BI** dashboard to monitor ETL batch processing in Snowflake, tracking the number of daily batch runs and their success and failure rates.

Data Science Intern, Danson Solutions, New Jersey

Jul 2021 - Sep 2021

- Developed a **Python** application integrating MySQL to analyze student performance data across quizzes, games, and interactive content in an e-learning site.
- Built dynamic visualizations using Tableau and **Matplotlib** to extract insights on learning engagement and content effectiveness by grade, subject, and school.
- Conducted exploratory data analysis (EDA) on educational datasets, uncovering patterns in performance based on user behavior, parental education, and school infrastructure.

TECHNICAL SKILLS AND CERTIFICATES

Programming Languages: Python, SQL, R, Java, C#, JavaScript, C++

Databases: SQL Server, MySQL, MongoDB, PostgreSQL

Tools and Platforms: Snowflake, Tableau, Power BI, Microsoft Azure, AWS (EC2, S3), MATLAB, Apache Kafka, Linux, Git Version Control, GitLab CI, Jupyter Notebook, Visual Studio Code, Docker, Selenium, Excel, LLM

Data science: Statistics, Machine Learning, Pandas, NumPy, Pytorch, Matplotlib, Scikit-learn

Certificates: R-Coding Essentials (Center for Analytics and Data Science - Miami University), Intermediate SQL (Datacamp), Intermediate Machine Learning (Kaggle), Microsoft Certified Azure Fundamentals

PROJECTS

Weather Comparison App

- Integrated Meteostat API to fetch historical weather data, enabling comparisons of current weather with the same day, week, or month from the previous year.
- Performed data analysis using **dplyr**, calculating daily, weekly, and monthly temperature differences.
- Created interactive visualizations with **ggplot2** to compare weather trends over different time frames (daily, weekly, monthly).
- Built web app using Shiny that allows users to select cities and date ranges and visualize temperature changes over time.

Identifying and Mining Accessibility Issues in Open-Source Projects on GitHub

- Analyzed accessibility-related issues in over 1000 popular GitHub repositories using the PyGithub library.
- Developed and implemented Python scripts to link accessibility issues to their corresponding fix commits.

Data Analysis and Visualization with R - Quality of Government Dataset

- Analyzed global socio-economic indicators (e.g., education, governance, GDP) by **filtering**, **grouping**, and **summarizing** data to uncover insights, such as education trends in high-GDP countries and correlations between proportional representation and political participation.
- Developed impactful visualizations using **ggplot2**, including scatter plots showcasing relationships like GDP vs. academic freedom and histograms illustrating distributions like academic freedom scores.

CIFAR-10 Image Classification using VGG architecture

- Built a 10-layer VGG-inspired CNN using **PyTorch** with batch normalization, dropout, and data augmentation, achieving 90% accuracy on the CIFAR-10 image classification dataset.

Flower Image Detection Android Application

- Fine-tuned a pre-trained MobileNetV2 model on a custom dataset of 15 flower categories, achieving an accuracy of 90%.
- Developed an Android mobile application using Java, integrating the trained model to enable users to capture images, classify flowers in real-time, and receive detailed descriptions.